

[Edit](#)**Guided**

Write Code with GitHub Copilot Suggestions

Challenge Overview

Understand the scenario

You are a Developer for Hexelo, an organization that needs to use GitHub Copilot for assisted development.

In this Challenge Lab, you will use Copilot to generate functions, create Python files, and update code. First, you will update GitHub Copilot in Visual Studio Code, and then you will accept Copilot suggestions for a function. Next, you will generate code by using a prompt and a command with Copilot. Finally, you will solve a programming challenge.

Navigating the Challenge Lab

Unable to process content from <https://raw.githubusercontent.com/LODSContent/Challenge-V3-Framework/main/Templates/Environments/None.md>

🔗 ► Quick tips for navigating the Challenge Lab instructions.

[New to Challenge Labs? Click here to learn more.](#)

Configure your environment and update GitHub Copilot in VS Code

Hints Enabled

No Yes

- Sign in to the virtual machine as **LabUser** by using **T** **Passw0rd!** as the password.
- Enter your GitHub Username, Email Address, Password, and Account URL into the following textboxes:

GitHub Username

GitHub Email Address

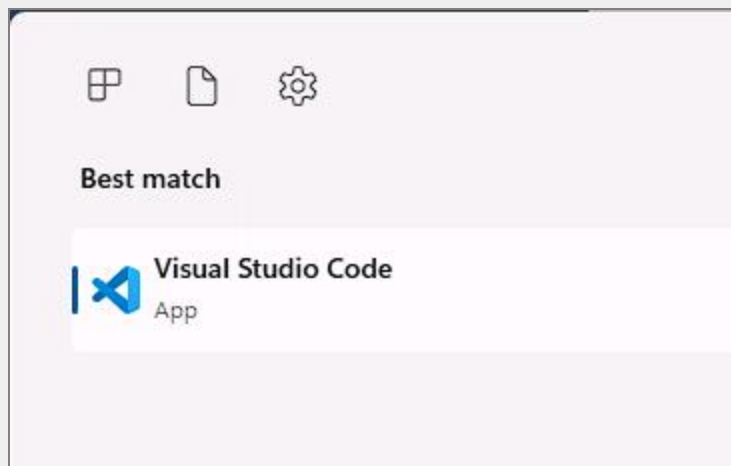
GitHub Password

Account URL

- Launch Visual Studio Code.

💡 Expand this hint for guidance on launching Visual Studio Code.

- On the Windows desktop, in Search, enter **Visual Studio Code**, and then select the app to run it.

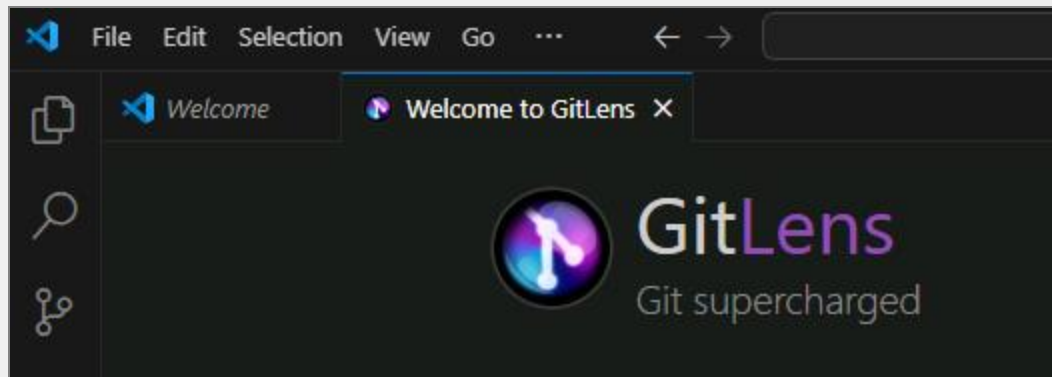


- If the Command prompt opens with a **Windows Subsystem** message, you may close it.

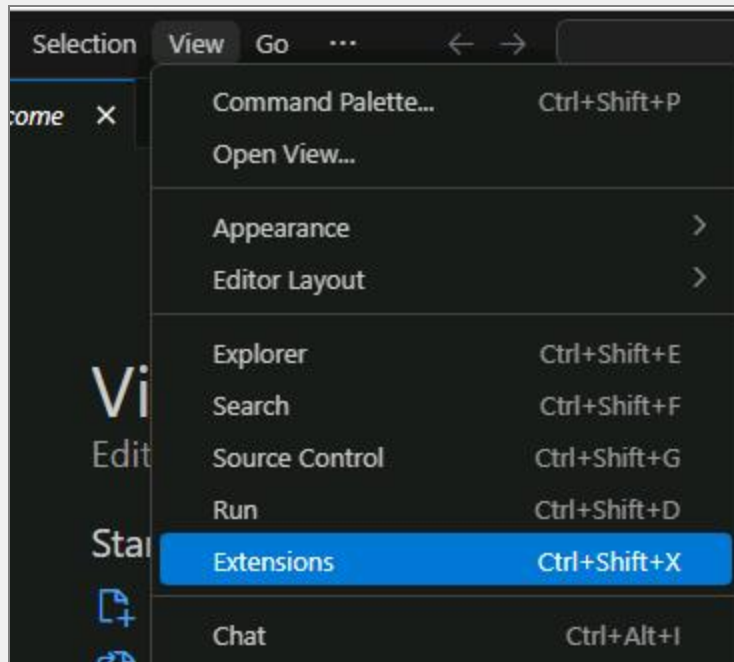
- Open the Extensions tab in Visual Studio Code.

💡 Expand this hint for guidance on opening the extensions tab.

- In the Visual Studio Code console, close the *Welcome to GitLens* tab.



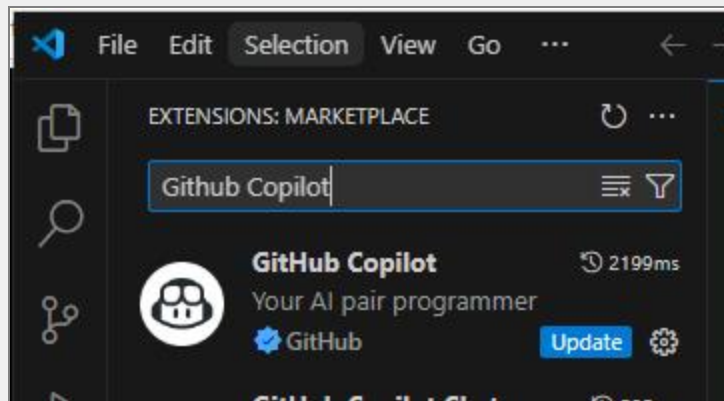
- On the Welcome tab, select **View**, and then select **Extensions** or select **Ctrl+Shift+X**.



- If necessary, install and update the [\[T\] GitHub Copilot](#) extension.

💡 Expand this hint for guidance on installing the GitHub Copilot extension.

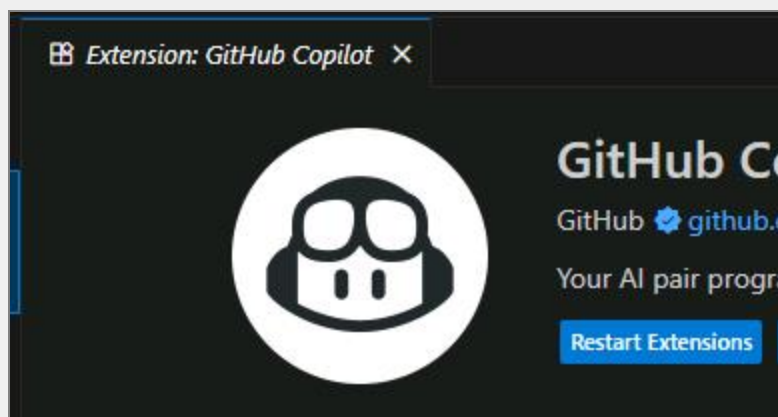
- On the Extensions pane, in *Search Extensions in Marketplace*, enter **GitHub Copilot**, and then select **GitHub Copilot**.



- On the *Extension: GitHub Copilot* tab, select **Update** (the version shown here may differ from the one in the current screenshot).



- Wait for the update to complete, and then select **Restart Extensions**.



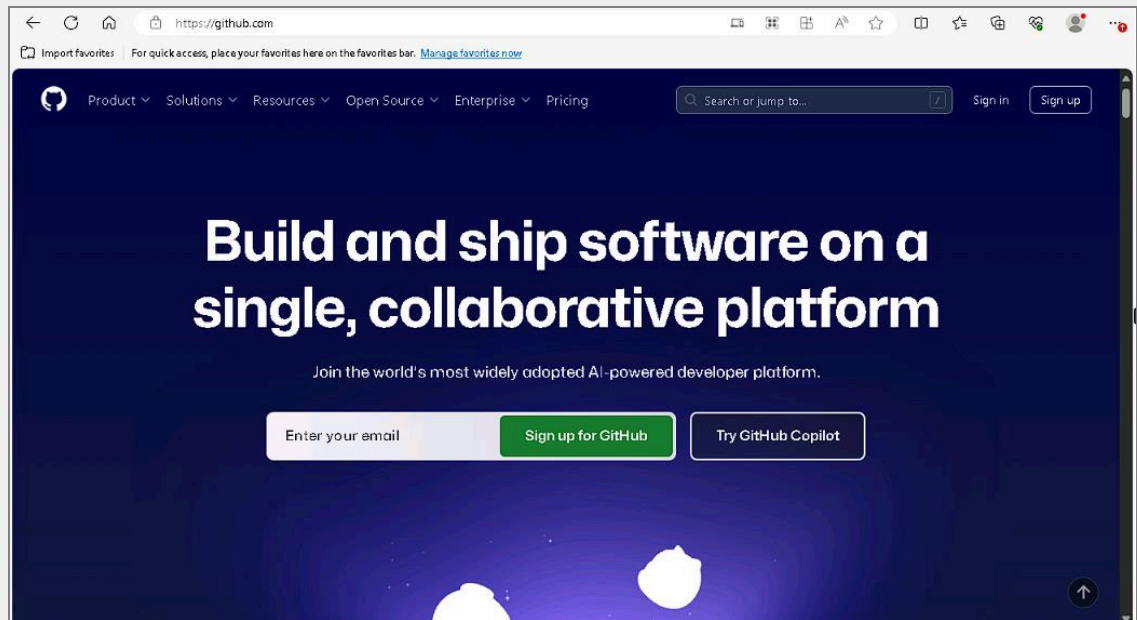


Want to learn more? Review the documentation on setting up the [GitHub Copilot extension](#) in Visual Studio Code.

- Open **Microsoft Edge**, go to <https://github.com>, and then sign in to your GitHub account as <GithubEmail> by using <GithubPW> as the password.

💡 Expand this hint for guidance on signing in to a GitHub account.

- On the Windows taskbar, in Search, enter **Edge**, and then select **Microsoft Edge** to open a new browser.
- In the Address bar, enter **T** <https://github.com>, and then press **Enter**.



- On the GitHub home page, in the upper right-hand corner, select **Sign in**.



- On the *Sign in to GitHub* page, in Username or email address, sign in as **T** [<GithubEmail>](#) by using **T** [<GithubPW>](#) as the password.

Device verification



Email

We just sent your authentication code via email to d*****@opseva.com. The code will expire at 5:44AM PDT.

Device Verification Code

Verify

Having trouble verifying via email?

- [Re-send the authentication code](#)
- [Try GitHub Mobile for simplified device verification](#)

If you'd like to require verification on every sign in, consider enabling [two-factor authentication](#) on your account.

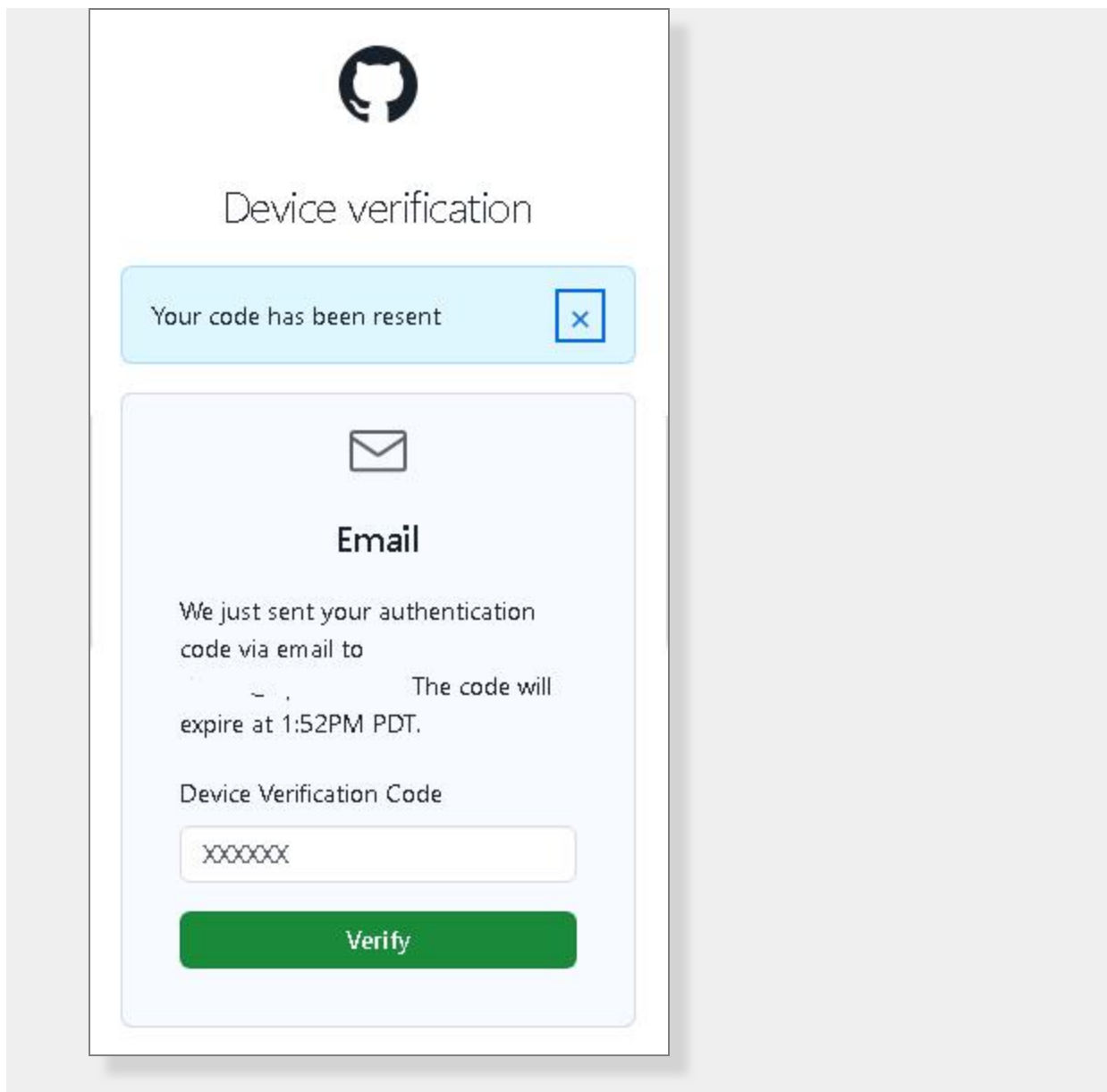
- On the Save password dialog, select **Never**.

 Saved passwords will be deleted on browser closure. [Change Setting.](#)

Got it

Never

- On the Device verification page, enter the **Device Verification Code** sent to the email address used to register the account.



 Want to learn more? Review the documentation on [signing in to a GitHub account](#).

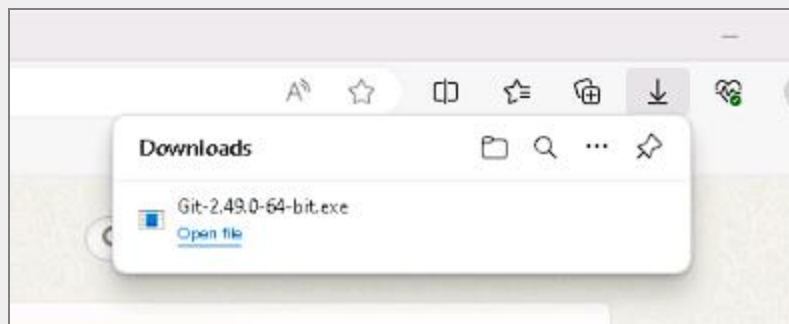
- If necessary, go to [T https://git-scm.com/download/win](https://git-scm.com/download/win), and then download and install **Git** by using the default options.

💡 Expand this hint for guidance on installing Git.

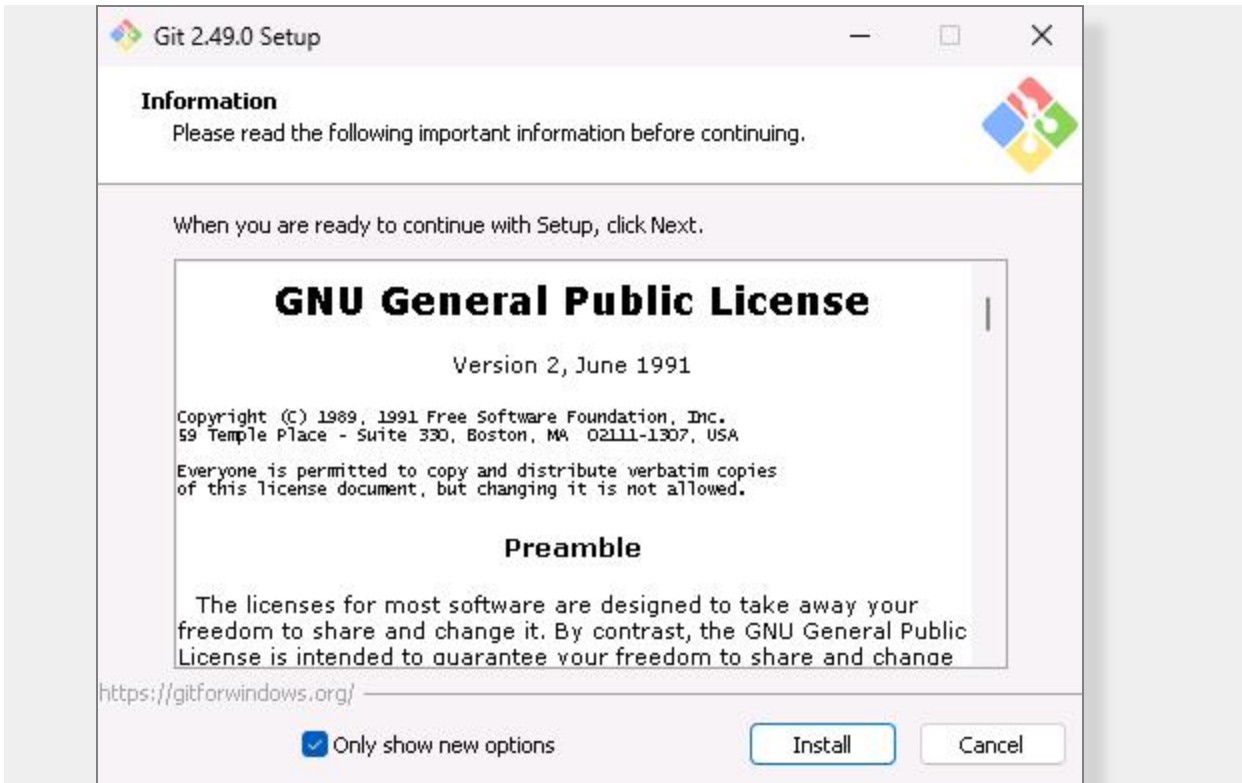
- In Microsoft Edge, open a new tab.
- In the Address bar, enter **T** <https://git-scm.com/download/win>, and then press **Enter**.
- On the **git** Download for Windows page, in Downloads, select **Click here to download**.



- Once the download completes, select **Open file**.



- On the *User Account Control - Do you want to allow this app to make changes to your device?* dialog, select **Yes**.
- On the Git Setup page, leave *Only show new options* selected, and then select **Install**.



- Once the Wizard is complete, select **Launch Git Bash**, deselect **View Release Notes**, and then select **Finish**.



Want to learn more? Review the documentation on [installing git](#).



Always verify Git is installed before running Git Bash commands by typing `git --version` in a terminal.

- Set your Git Identity by using the `git config` command and using `<Name>` as the **username** and `<GithubEmail>` as the **email address**.



Expand this hint for guidance on setting your Git Identity.

- In the Git Bash Terminal, run the following commands to set your Git identity-press **Enter** on your keyboard after entering a command.

```
git config --global user.name "<Name>"
```

```
git config --global user.email "<GithubEmail>"
```

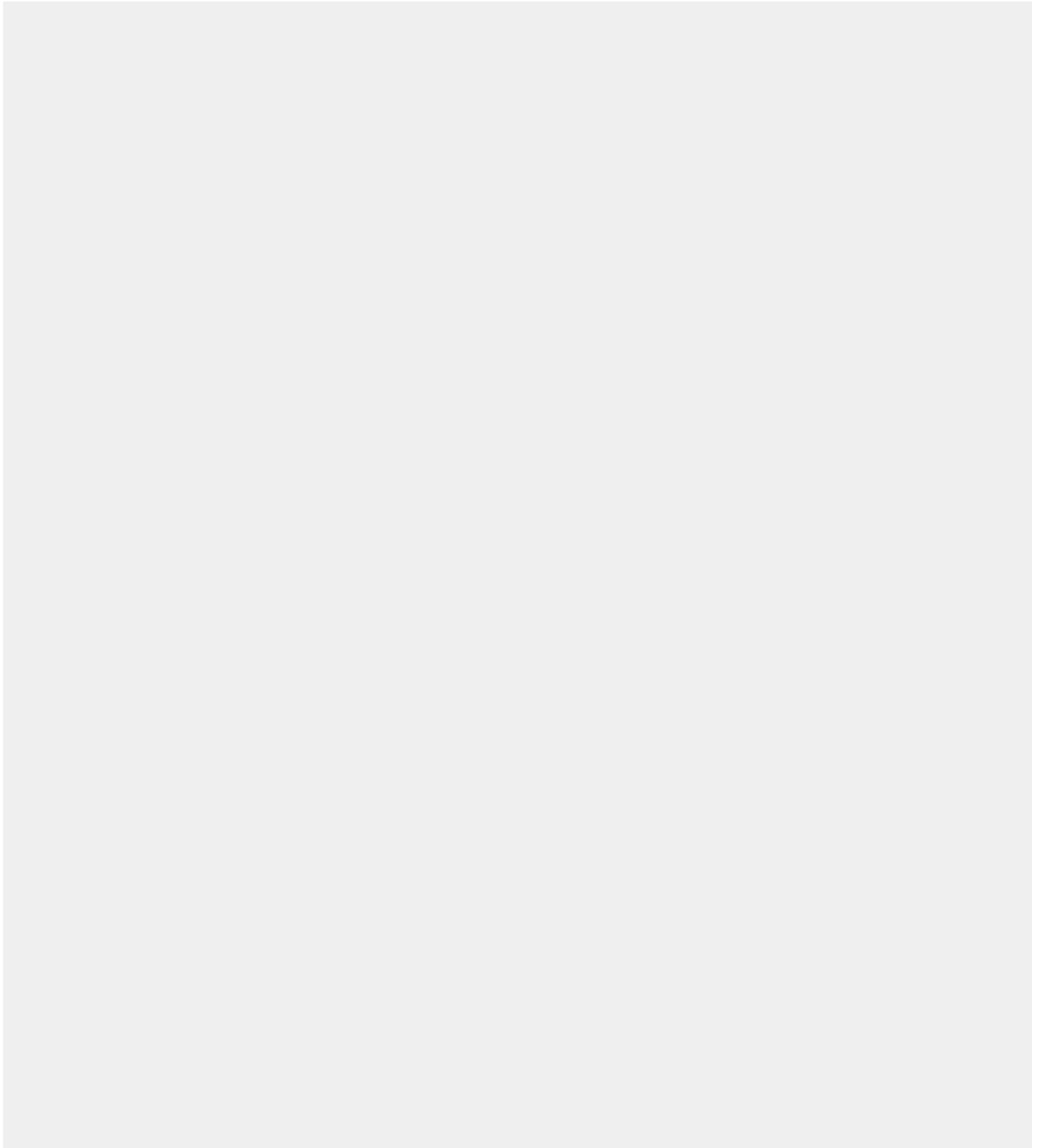


Want to learn more? Review the documentation on [setting your Git Identity](#).

Check your work



Verify



Generate code from a prompt and then modify a GitHub Copilot suggestion

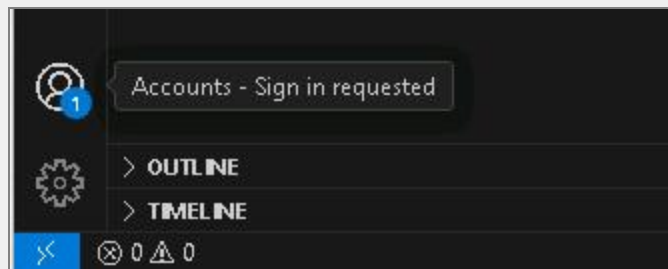
Hints Enabled

No Yes

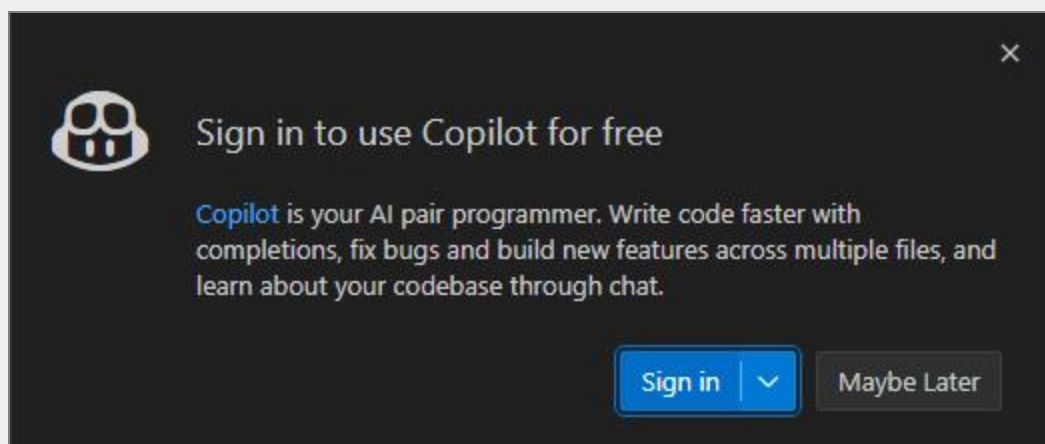
- Sign in to Copilot as T <GithubEmail> by using T <GithubPW> as the password.

 Expand this hint for guidance on signing in to Copilot. 

- Select the **Accounts - Sign in requested** icon in the lower left-hand corner, and then select **Sign in with GitHub to use GitHub Copilot**.



- On the *Sign in to use Copilot for free* dialog, select **Sign in**, or select **Continue**.



- On the *Authorize page*, select **Authorize Visual-Studio-Code**.
- On the *This site is trying to open Visual Studio Code* dialog, select **Open**.



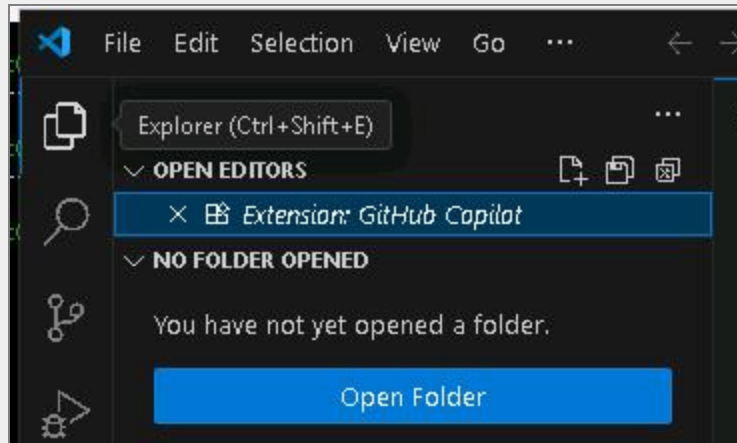
Want to learn more? Review the documentation on [signing in to Copilot](#).

- Using the following text, create a new file in VS Code using Copilot that will write a function that reads a file and returns the number of lines, and then save it as `file_utils.py` in the `D:\LabFiles` folder.

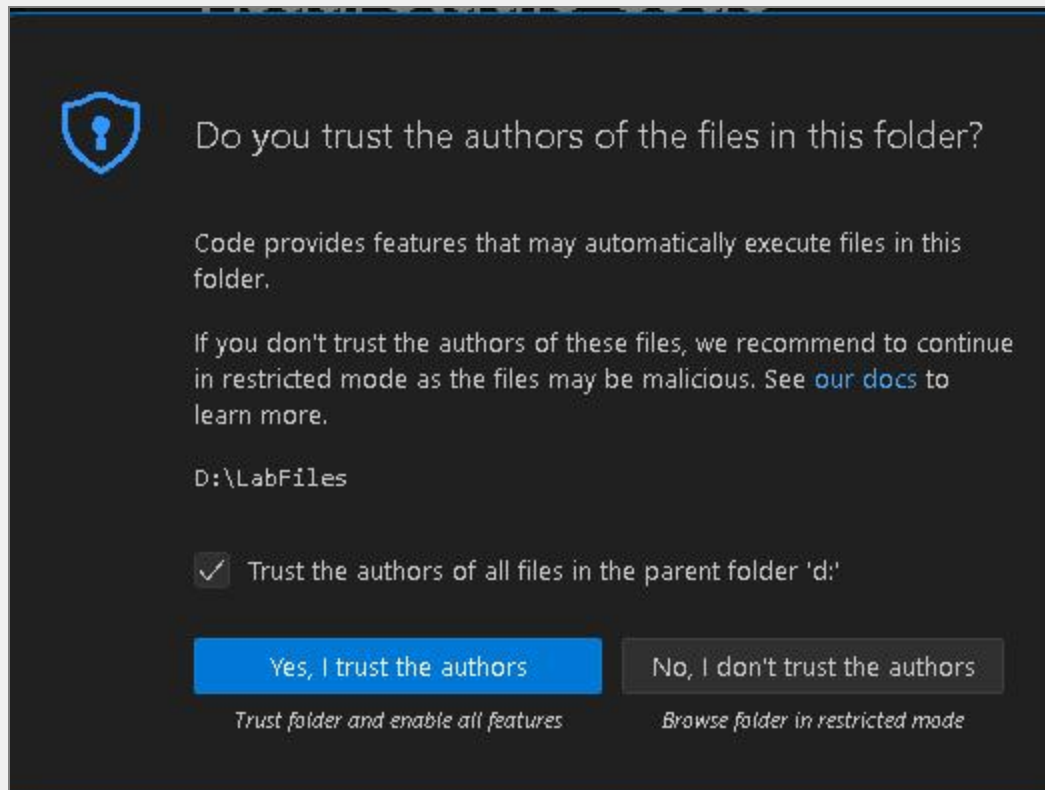
`Write a function that reads a file and returns the number of lines`

💡 Expand this hint for guidance on creating a Python script in VS Code using Copilot. ^

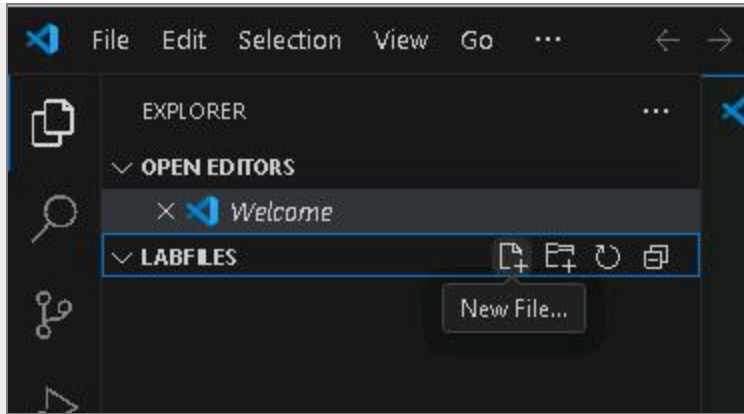
- In the Visual Studio Code console, select the **Explorer** icon.



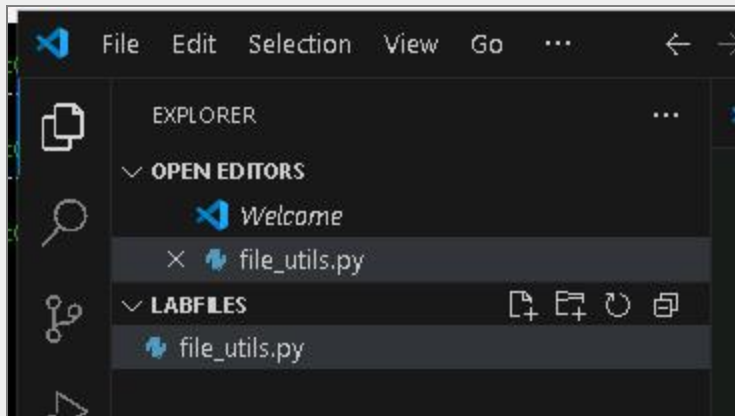
- In the Explorer, in *NO FOLDER OPENED*, select **Open Folder**, navigate to the T **D:\LabFiles** folder, and then select **Select folder**.
- If the Command prompt opens with a **Windows Subsystem** message, you may close it.
- On the *Do you trust the authors of the files in this folder?* dialog, enable the checkbox, and then select **Yes, I trust the authors**.



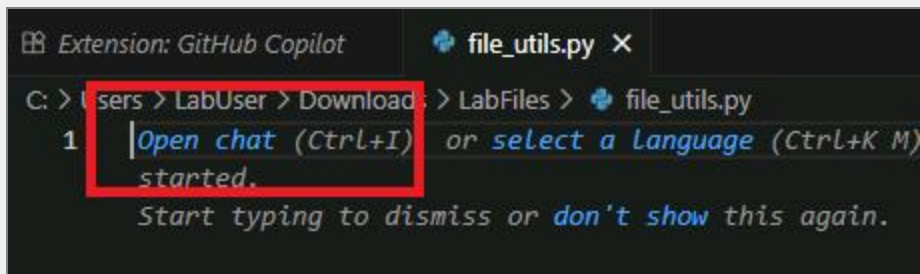
- In the Explorer, in the *LabFiles* folder, select **New file**.



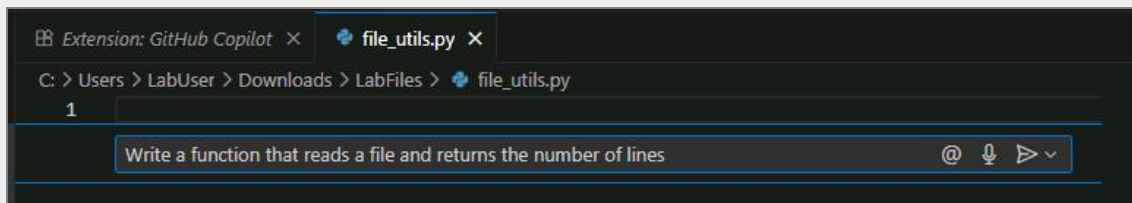
- Enter `file_utils.py` and press **Enter**.



- On the `file_utils.py` tab, at the blinking prompt, select **Open chat**, or select **Ctrl+I** to open Chat.

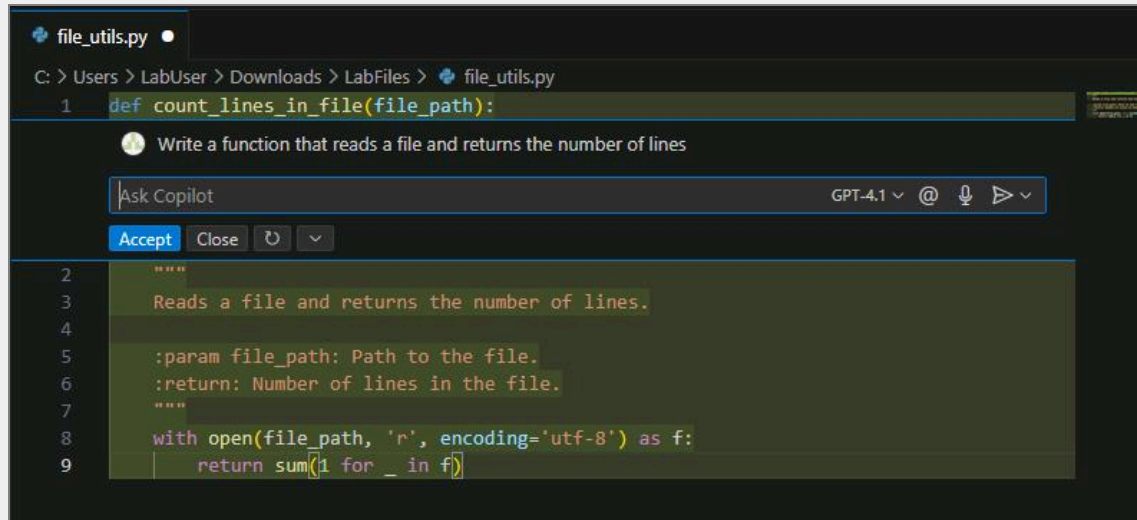


- In the Ask Copilot prompt, enter `Write a function that reads a file and returns the number of lines`, and then select **Send and Dispatch (Enter)**.



- If necessary, on the *Start using Copilot for free* dialog, select **Use Copilot**.

- Review Copilot's suggestions, and then select **Accept**.



The screenshot shows the Visual Studio Code editor with a file named `file_utils.py` open. The file path is `C:\Users\LabUser\Downloads\LabFiles\file_utils.py`. The current line of code is `def count_lines_in_file(file_path):`. A Copilot suggestion is displayed, showing a function that reads a file and returns the number of lines. The suggestion includes a docstring, a parameter description, and the implementation using `open` and `sum`. The suggestion is presented in a dark-themed box with a light background for the code. Below the suggestion, there are buttons for `Accept`, `Close`, and a refresh icon. The `Accept` button is highlighted in blue.

```
1 def count_lines_in_file(file_path):  
2     """  
3     Reads a file and returns the number of lines.  
4     """  
5     :param file_path: Path to the file.  
6     :return: Number of lines in the file.  
7     """  
8     with open(file_path, 'r', encoding='utf-8') as f:  
9         return sum(1 for _ in f)
```

 Want to learn more? Review the documentation on [creating a Python script in VS Code](#).

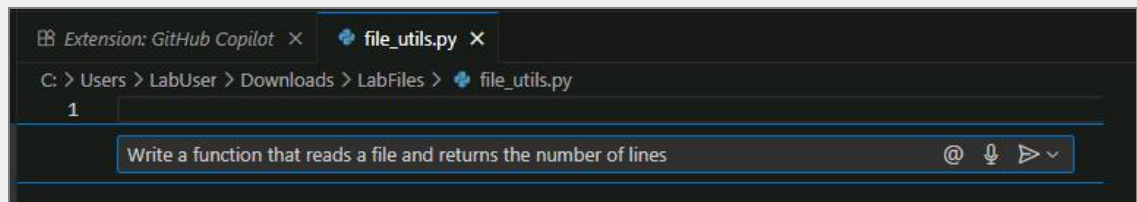
 Wait 1-2 seconds for Copilot to suggest a full function.

- Using the following text, update the file in VS Code using Copilot to write a function that counts the number of words in a file, and then save it as `file_utils.py` in the `D:\LabFiles` folder.

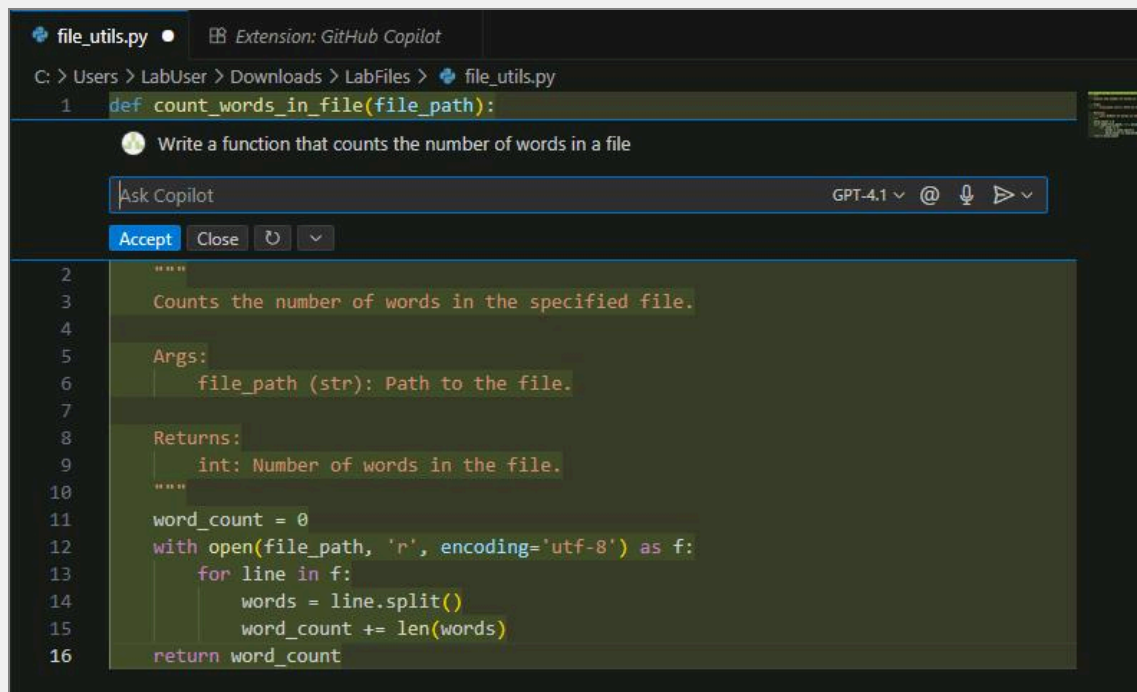
`Write a function that counts the number of words in a file`

💡 Expand this hint for guidance on modifying a Python script in VS Code.

- Select **Ctrl+A**, and then delete the currently suggested script.
- On the `file_utils.py` tab, at the blinking prompt, select **Open chat**, or select **Ctrl+I** to open Chat.
- In the *Ask Copilot* prompt, enter `T` Write a function that counts the number of words in a file, and then select **Send and Dispatch (Enter)**.



- Review Copilot's suggestions, and then select **Accept**.



- Select **File | Save**.

Check your work

Verify

Generate code from a prompt and a command using GitHub Copilot

Hints Enabled

No Yes

- Using the following text, create a new file in VS Code that will write a function that returns a list of all filenames in a directory, and then save it as `file_utils2.py` in the `D:\LabFiles` folder.

`Write a function that returns a list of all filenames in a directory`

 Expand this hint for guidance on creating a Python script in VS Code using Copilot. 

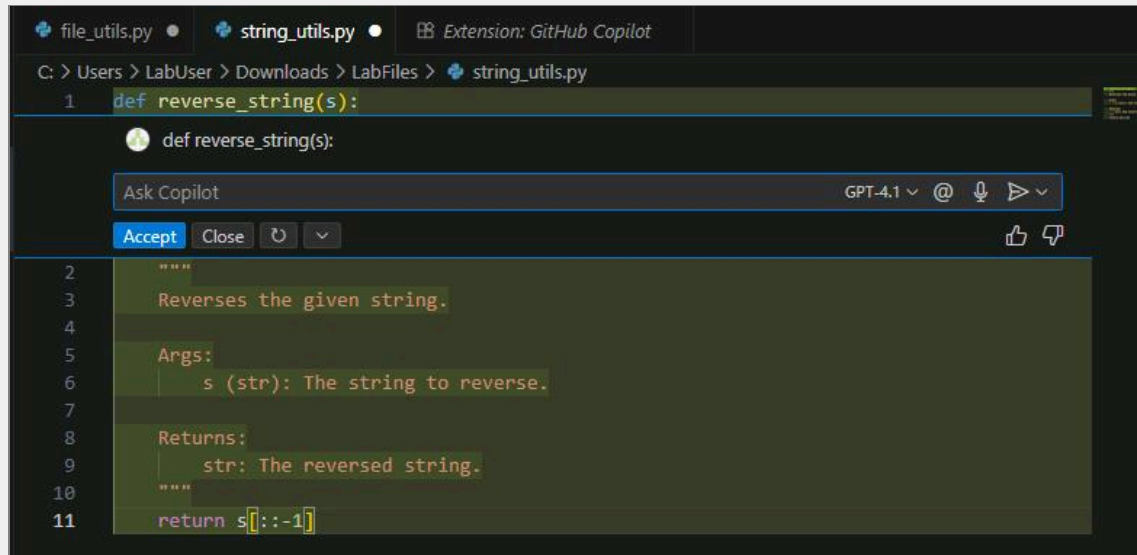
- In the Visual Studio Code console, in the **Explorer** panel, select the **D:\LabFiles** folder, and then select **New file**.
- Enter `file_utils2.py` and press **Enter**.
- On the `file_utils2.py` tab, at the blinking prompt, select **Open chat**, or select **Ctrl+I** to open Chat.
- In the *Ask Copilot* prompt, enter `Write a function that returns a list of all filenames in a directory`, and then select **Send and Dispatch (Enter)**.
- Review Copilot's suggestions, and then select **Accept**.
- Select **File | Save**.

- Using the following command, create a new file in VS Code, and then save it as `string_utils.py` in the `D:\LabFiles` folder.

`def reverse_string(s):`

💡 Expand this hint for guidance on creating a Python script in VS Code using Copilot. ^

- In the Visual Studio Code console, in the **Explorer** panel, select the **D:\LabFiles** folder, and then select **New file**.
- Enter `string_utils.py` and press **Enter**.
- On the `string_utils.py` tab, at the blinking prompt, select **Open chat**, or select **Ctrl+I** to open Chat.
- In the *Ask Copilot* prompt, enter `def reverse_string(s):`.



```
1 def reverse_string(s):
2     """
3     Reverses the given string.
4
5     Args:
6         s (str): The string to reverse.
7
8     Returns:
9         str: The reversed string.
10    """
11    return s[::-1]
```

- Review Copilot's suggestions, and then select **Accept**.
- Select **File | Save**.

Check your work

Verify

Solve a programming challenge

Hints Enabled

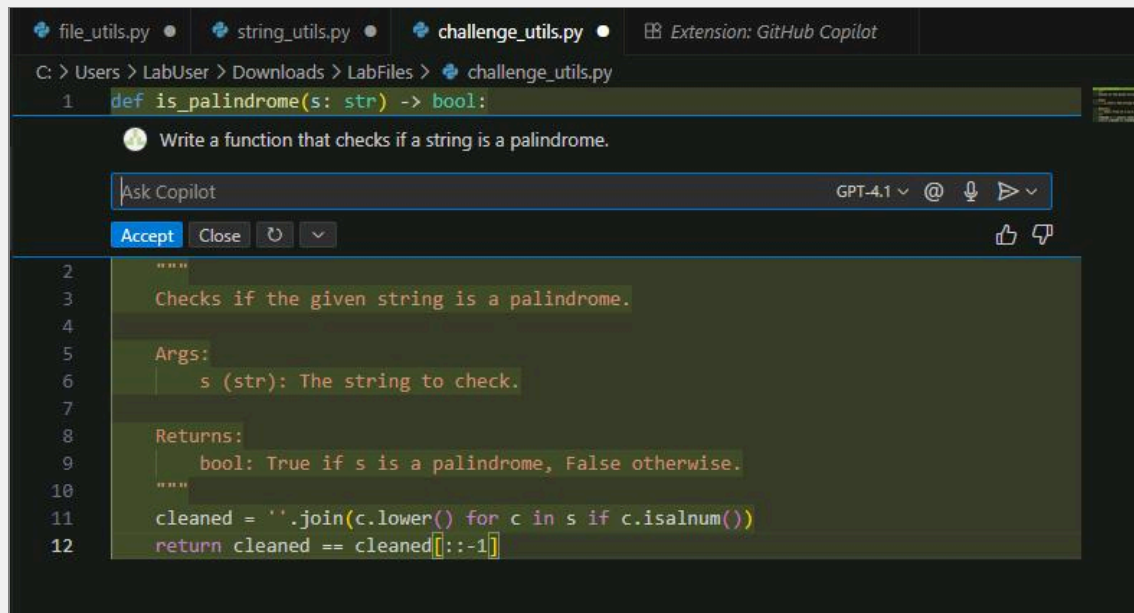
No Yes

- Using the following text, create a new file in VS Code that will write a function that returns a list of all filenames in a directory, and then save it as `T challenge_utils.py` in the `T D:\LabFiles` folder.

`T` Write a function that checks if a string is a palindrome

💡 Expand this hint for guidance on creating a Python script in VS Code using Copilot. ^

- In the Visual Studio Code console, in the **Explorer** panel, select the **D:\LabFiles** folder, and then select **New file**.
- Enter `challenge_utils.py` and press **Enter**.
- On the `challenge_utils.py` tab, at the blinking prompt, select **Open chat**, or select **Ctrl+I** to open Chat.
- In the *Ask Copilot* prompt, enter `Write a function that checks if a string is a palindrome`, and then press **Enter**.



```
file_utils.py • string_utils.py • challenge_utils.py • Extension: GitHub Copilot
C: > Users > LabUser > Downloads > LabFiles > challenge_utils.py
1 def is_palindrome(s: str) -> bool:
    Write a function that checks if a string is a palindrome.
    Ask Copilot GPT-4.1 @
    Accept Close
2 """
3 Checks if the given string is a palindrome.
4
5 Args:
6     s (str): The string to check.
7
8 Returns:
9     bool: True if s is a palindrome, False otherwise.
10 """
11 cleaned = ''.join(c.lower() for c in s if c.isalnum())
12 return cleaned == cleaned[::-1]
```

- Review Copilot's suggestions, and then select **Accept**.
- Select **File | Save**.

- In VS Code, in `challenge_utils.py`, test to determine if `print(is_palindrome("racecar"))` is a palindrome.

💡 Expand this hint for guidance on solving a programming challenge. ^

- In the VS Code console, in the *challenge_utils.py* script, after the **return cleaned** function, enter `T print(is_palindrome("racecar"))`. You should see the value **# True** appear.

```
cleaned = ''.join(c.lower() for c in s if c.isalnum())
return cleaned == cleaned[::-1]

print(is_palindrome("racecar")) # True
```

- Select **tab** to accept.

- In VS Code, in *challenge_utils.py*, test to determine if `T print(is_palindrome("hello"))` is a palindrome.

```
11 cleaned = ''.join(c.lower() for c in s if c.isalnum())
12 return cleaned == cleaned[::-1]
13
14 print(is_palindrome("racecar")) # True
   print(is_palindrome("hello")) # False
```

- Save your work by selecting **File | Save**.

Check your work

Verify

Summary

Congratulations, you have completed the **Write Code with GitHub Copilot Suggestions** Challenge Lab.

You have accomplished the following:

- Configured your environment and updated GitHub Copilot in VS Code.
- Generated code from a prompt and then modified a GitHub Copilot suggestion.
- Generated code from a prompt and a command using GitHub Copilot.
- Solved a programming challenge.

Ending your lab

To ensure your lab is recorded as complete, select **Submit** or **End** below. Exiting [X] the lab will result in an incomplete status.

 Once you select **Submit** or **End**, you will not be able to return to this Challenge Lab.

Your feedback is important!

As you end your Challenge Lab, please take a few minutes to complete the short survey that will appear in the next window. Alternatively, you may provide your feedback directly to [Challenge Labs feedback](#).

Looking for your next Challenge Lab?

Below are recommended Challenge Labs to try next.

- Set Up Git and Connect to GitHub [Prerequisite Lab]
- Track Local Changes and Manage Commits [Guided]
- Create Branches and Merge Changes Safely [Guided]
- Collaborate Remotely and Use Pull Requests [Guided]
- Build a Real Project by Using Git Workflows [Advanced]
- Write Code with GitHub Copilot Suggestions [Guided]

- Control Git Behavior with Config and Ignore Files [Guided]
- Rebase, Stash, and Apply Commits Precisely [Guided]
- Manage Tasks by Using GitHub Issues and Boards [Guided]
- Automate Testing and CI with GitHub Actions [Guided]
- Review and Refactor Code with Copilot [Guided]
- Implement Release Management and Versioning in GitHub [Expert]

